Probability

Random experiment

An experiment that can result in different outcomes, even though it is repeated in the same manner every time, is called a random experiment

EXAMPLE 1

1. If we toss a coin, the result of the experiment is that it will either come up "tails," symbolized by T , or "heads," symbolized by H , i.e., one of the elements of the set {H, T}
2. If we toss a die, the result of the experiment is that it will come up with one of the numbers in the set {1, 2, 3, 4, 5, 6}

3. If we toss a coin twice, there are four results possible, as indicated by {HH, HT, TH, TT}, i.e., both heads, heads on first and tails on second, etc.

Sample Spaces

A set S that consists of all possible outcomes of a random experiment is called a sample space, and each outcome is called a sample point.

Example:

If we toss a die, one sample space, or set of all possible outcomes, is given by {1, 2, 3, 4, 5, 6}.

Events

An event is a subset A of the sample space S.

Example

If we toss a coin twice, the event that only one head comes up is the subset of the sample space that consists of points (H, T) and (T, H)

Set operations

If A and B are events, then

- $\boldsymbol{A} \cup \boldsymbol{B}$ Is the event "either A or B or both." $A \cup B$ is called the **union** of A and B.

- $\boldsymbol{A} \cap \boldsymbol{B}$ Is the event "both A and B." $A \cap B$ is called the **intersection** of A and B.

- A is the event "**not A**." $\boldsymbol{A'}$ is called the **complement** of A.

Set Notation

Sets of Data

Universal Set → $\mathbf{S}$ = {Numbers 1 to 20 inclusive}

Elements

$\mathbf{A}$ = {Square Numbers} = {1, 4, 9, 16}

$\mathbf{B}$ = {Multiples of 4} = {4, 8, 12, 16, 20}

What is…

A ∪ B

Union: the combination of **A** and **B**.

$A \cup B = \{1, 4, 8, 9, 12, 16, 20\}$

A ∩ B

Intersection: The overlap of **A** and **B**.

$A \cap B = \{4, 16\}$

A'

Complement: Not in **A**.

$A' = \{2, 3, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 17, 18, 19, 20\}$

Venn Diagrams & Set Notation

S = {Numbers 1 to 9 inclusive}

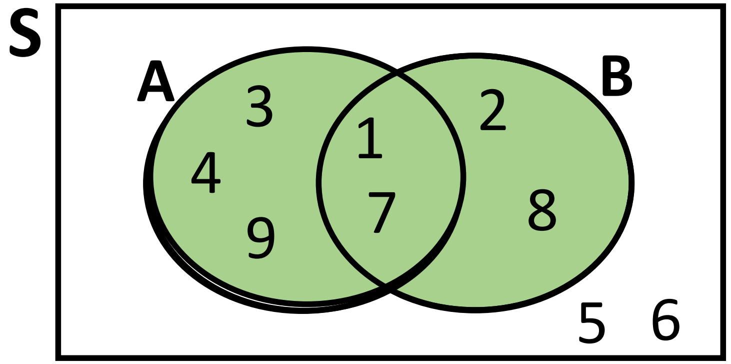
A = {1, 3, 4, 7, 9}

B = {1, 2, 7, 8}

Union

A ∪ B = {1, 2, 3, 4, 7, 8, 9}

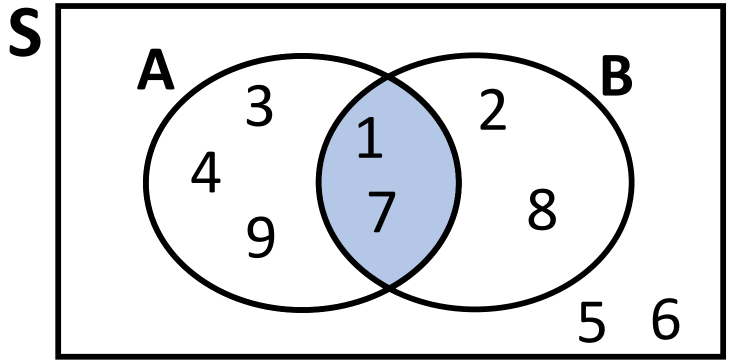
The combination of **A** and **B**.



Intersection

A ∩ B = {1, 7}

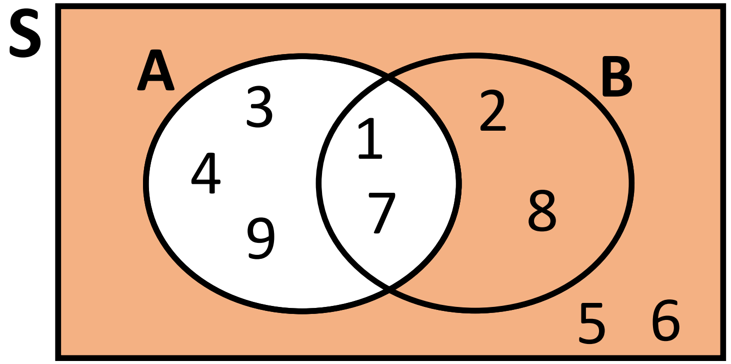
The overlap of **A** and **B**.



Complement

A’ = {2, 5, 6, 8}

Not in **A**.



Using totals in a Venn Diagram…

Union

A ∪ B

The combination of **A** and **B**.

Intersection

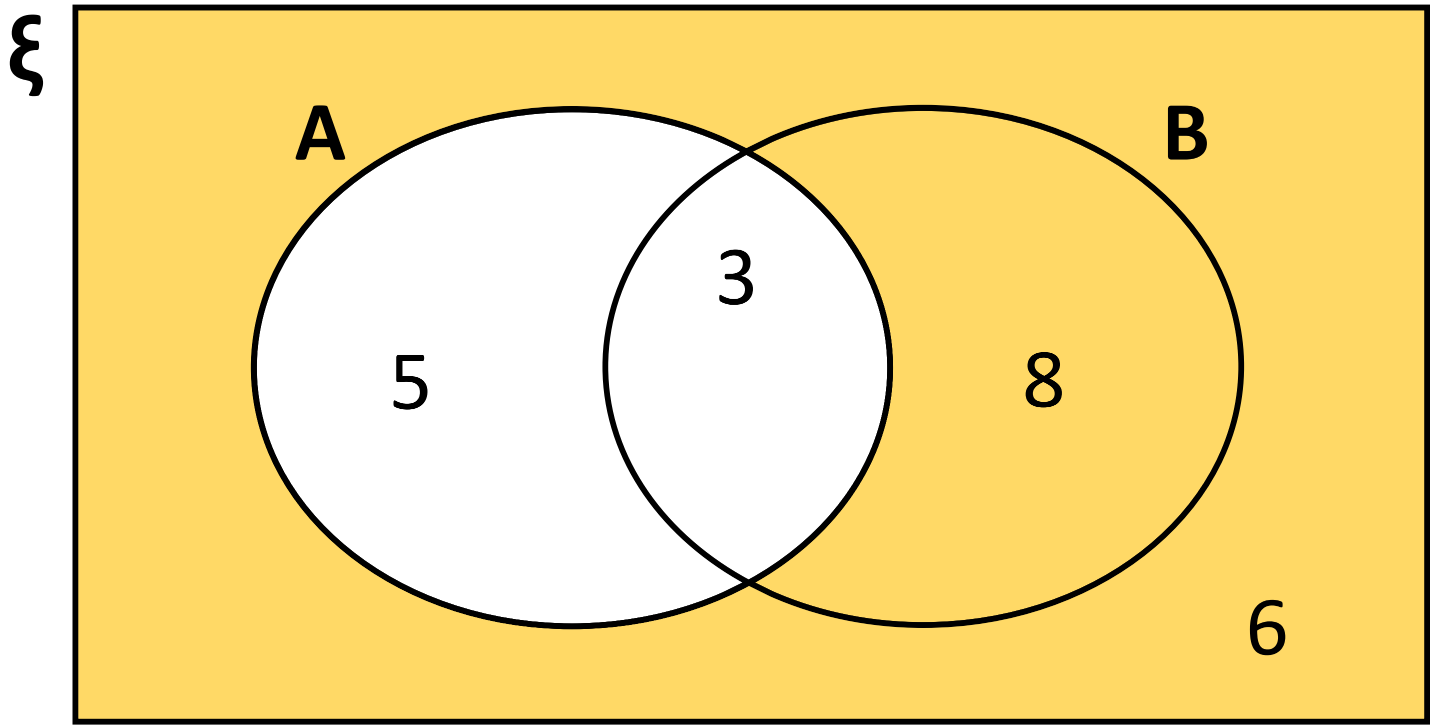
A ∩ B

The overlap of **A** and **B**.

Complement

A′

Not in **A**.



Describe & total the shaded area

A′ = 14

Using totals in a Venn Diagram…

Union

A ∪ B

The combination of **A** and **B**.

Intersection

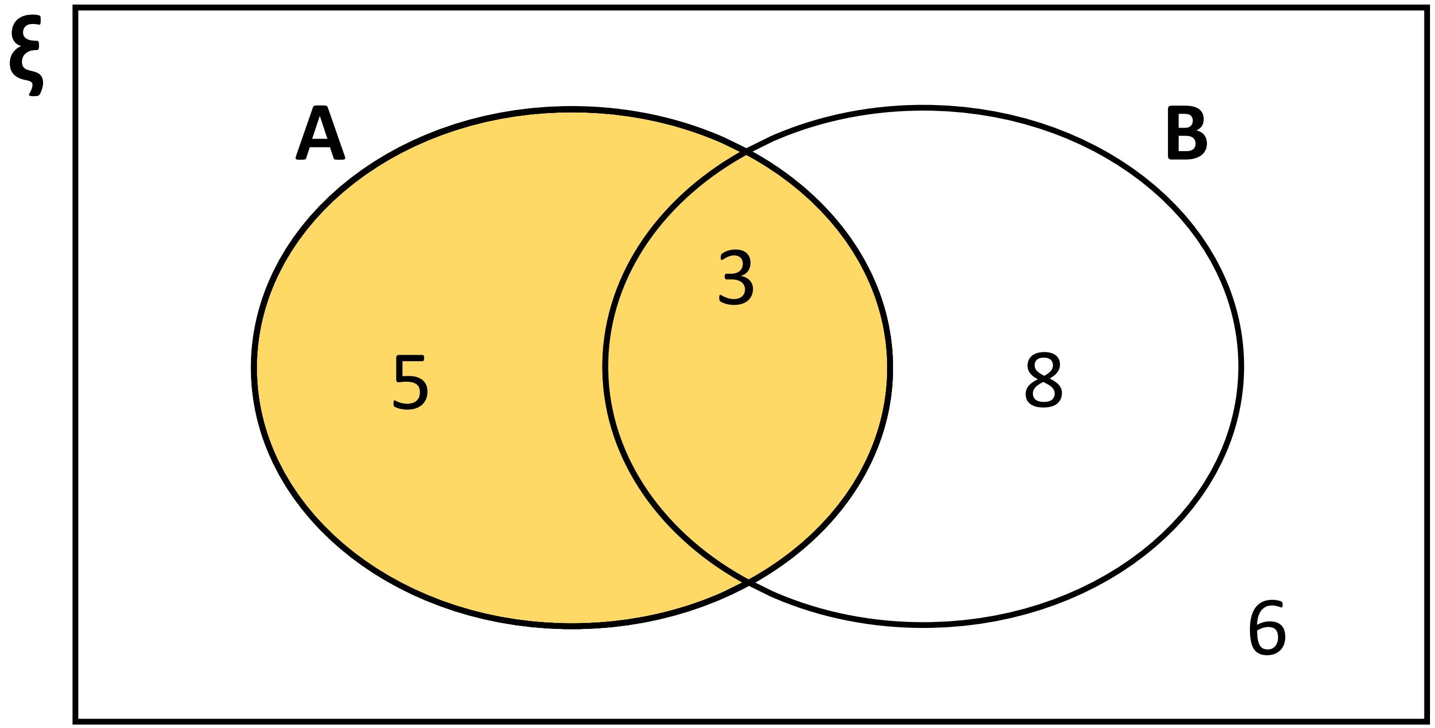
A ∩ B

The overlap of **A** and **B**.

Complement

A’

Not in **A**.



Describe & total the shaded area

A = 8

Using totals in a Venn Diagram…

Union

A ∪ B

The combination
of **A** and **B**.

Intersection

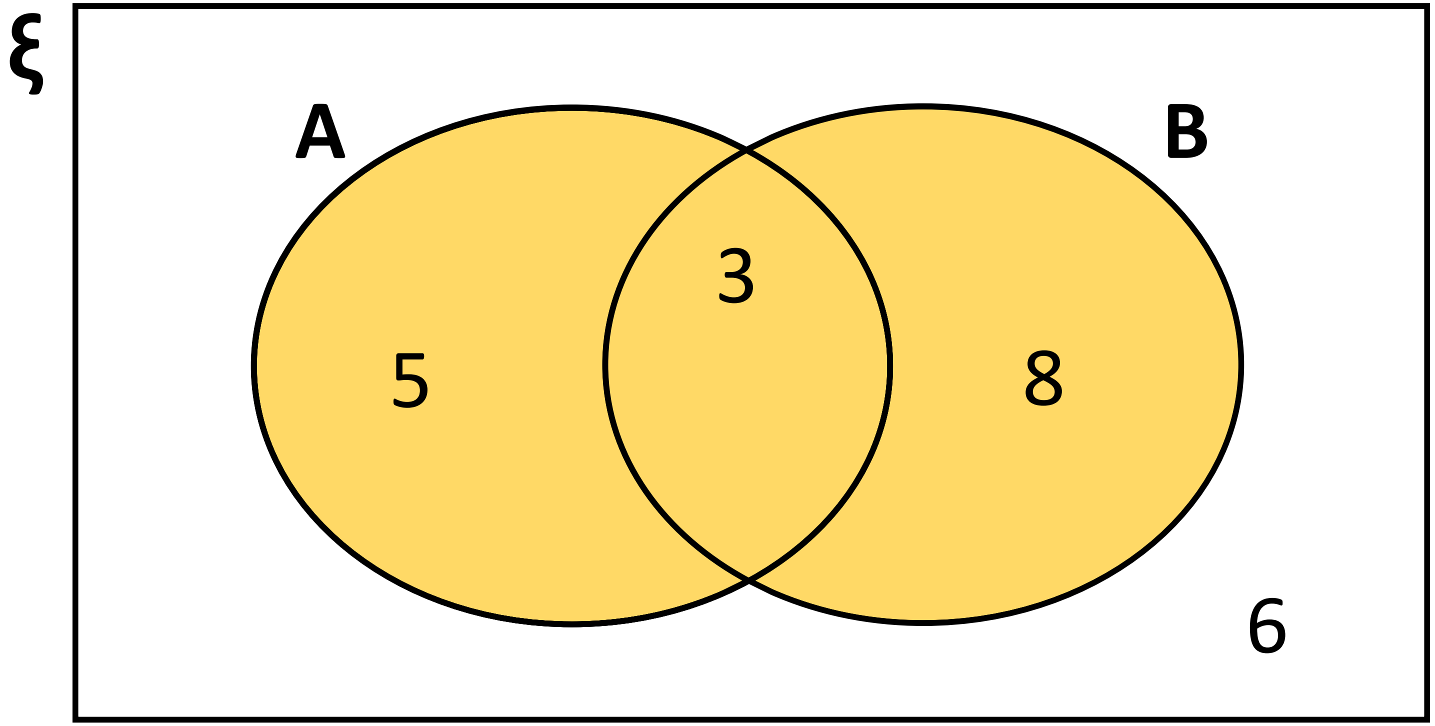
A ∩ B

The overlap
of **A** and **B**.

Complement

A’

Not in **A**.



Describe & total the shaded area

A ∪ B = 16

Using totals in a Venn Diagram…

Union

A ∪ B

The combination of **A** and **B**.

Intersection

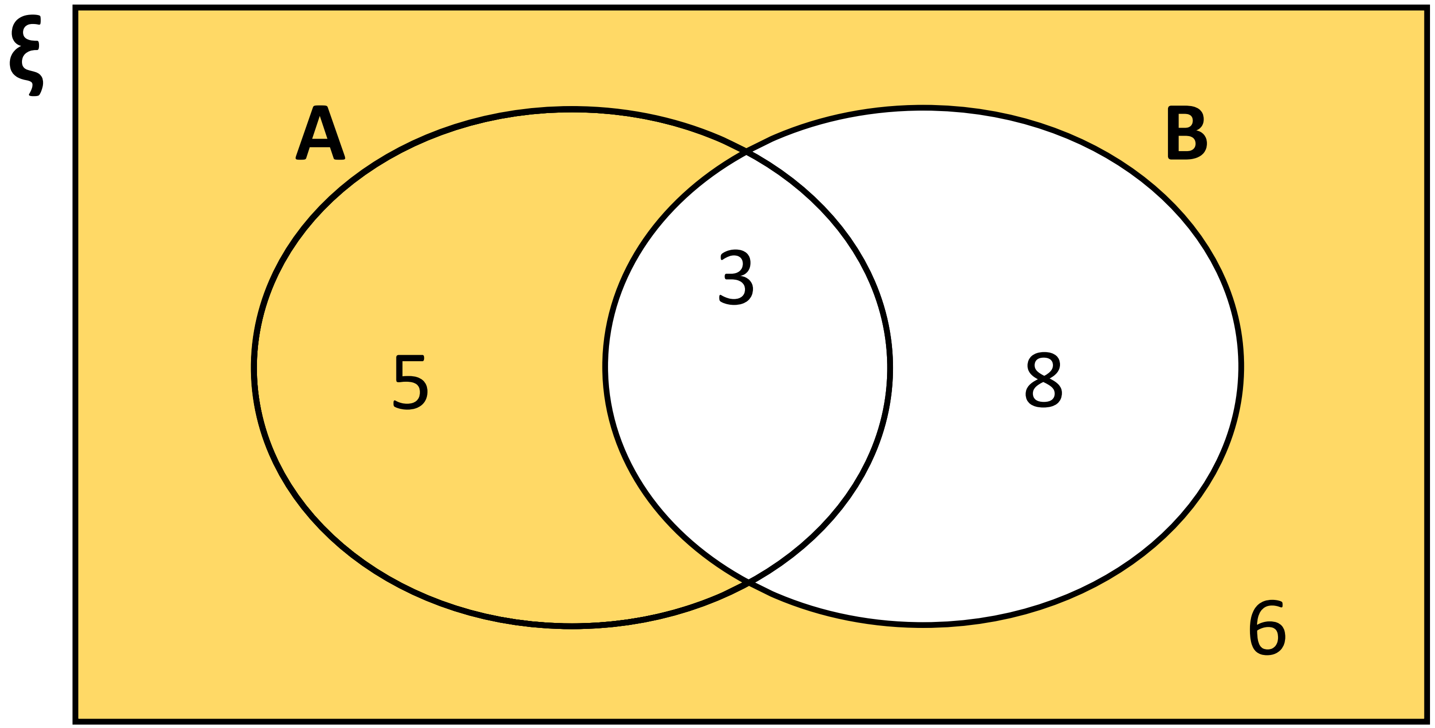
A ∩ B

The overlap of **A** and **B**.

Complement

A’

Not in **A**.



Describe & total the shaded area

B’ = 11

Using totals in a Venn Diagram…

Union

A ∪ B

The combination of **A** and **B**.

Intersection

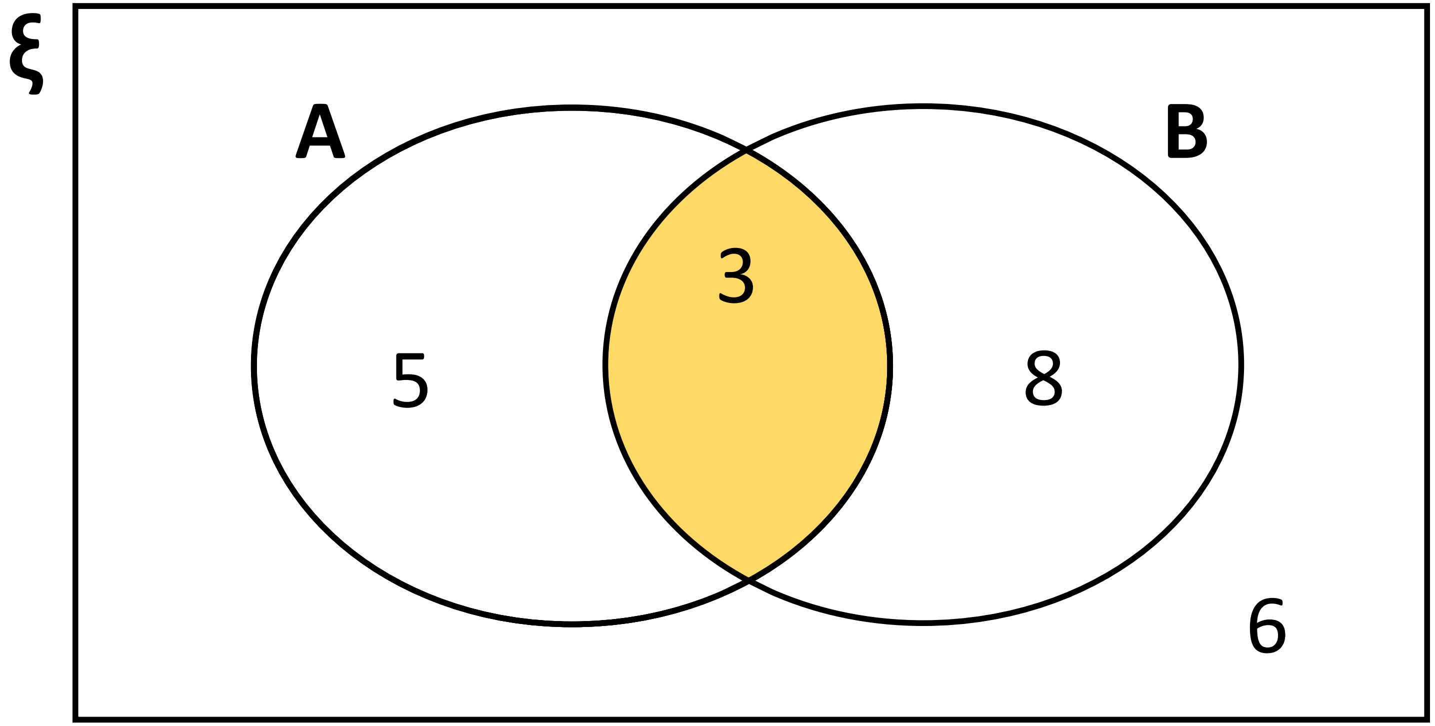
A ∩ B

The overlap of **A** and **B**.

Complement

A’

Not in **A**.



Describe & total the shaded area

A ∩ B = 3

Using totals in a Venn Diagram…

Union

A ∪ B

The combination of **A** and **B**.

Intersection

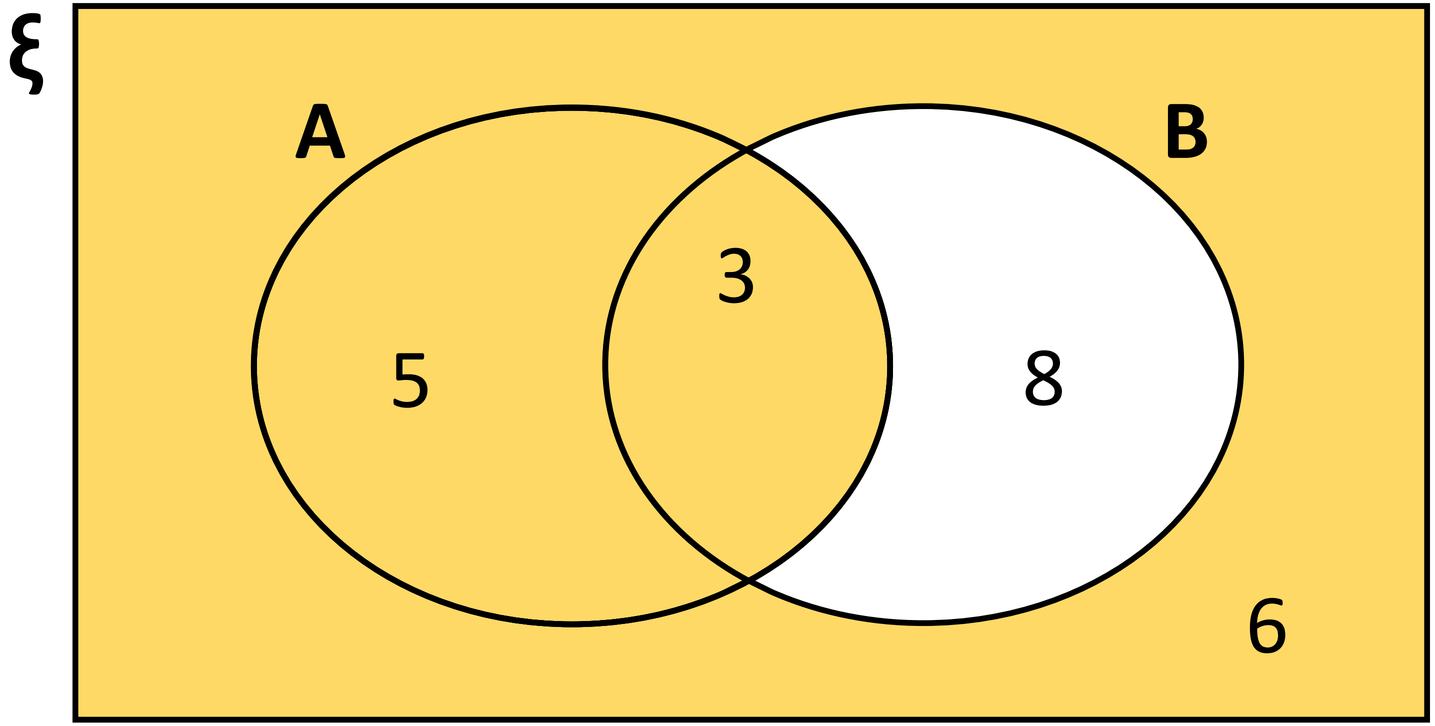
A ∩ B

The overlap of **A** and **B**.

Complement

A'

Not in **A**.



Describe & total the shaded area

A ∪ B' = 14

Using totals in a Venn Diagram…

Union

A ∪ B

The combination of **A** and **B**.

Intersection

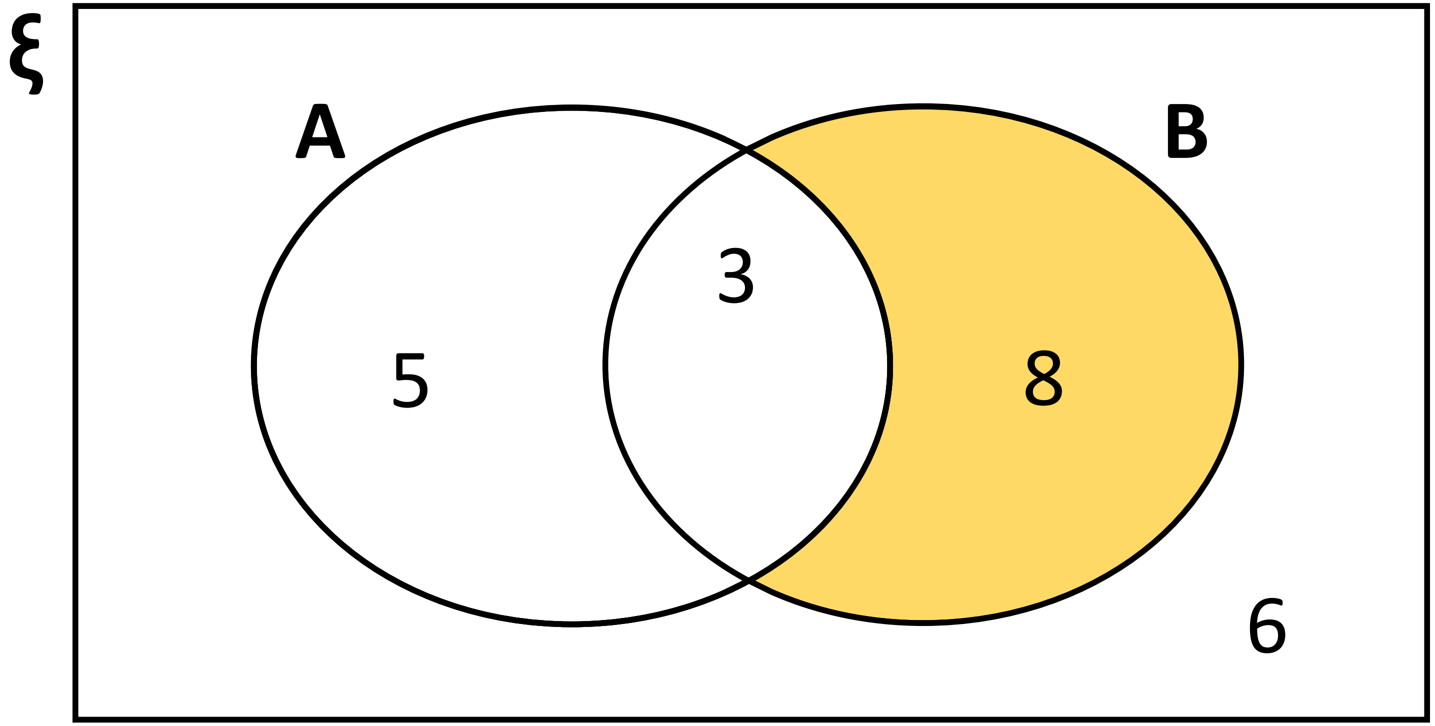
A ∩ B

The overlap of **A** and **B**.

Complement

A’

Not in **A**.



Describe & total the shaded area

A’ ∩ B = 8

Using totals in a Venn Diagram…

Union

A ∪ B

The combination of **A** and **B**.

Intersection

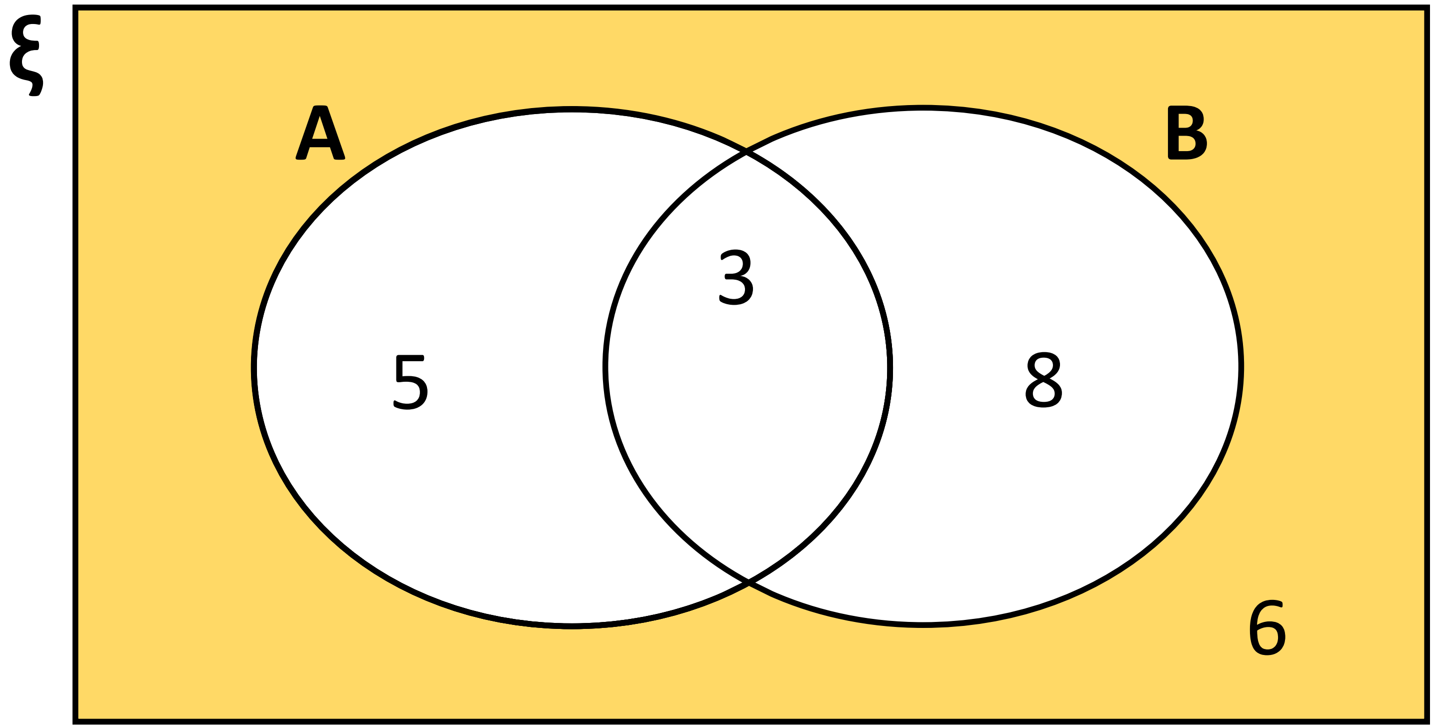
A ∩ B

The overlap of **A** and **B**.

Complement

A’

Not in **A**.



Describe & total the shaded area

A’∩B’ = 6

Union

A ∪ B

The combination of **A** and **B**.

Intersection

A ∩ B

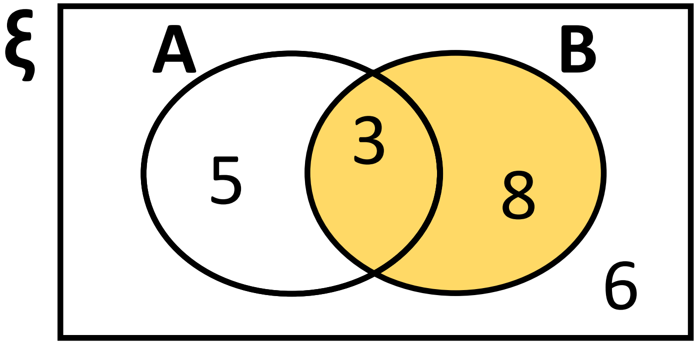
The overlap of **A** and **B**.

Complement

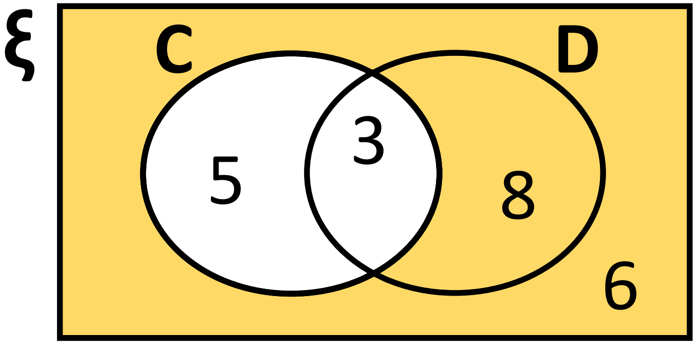
A′

Not in **A**.

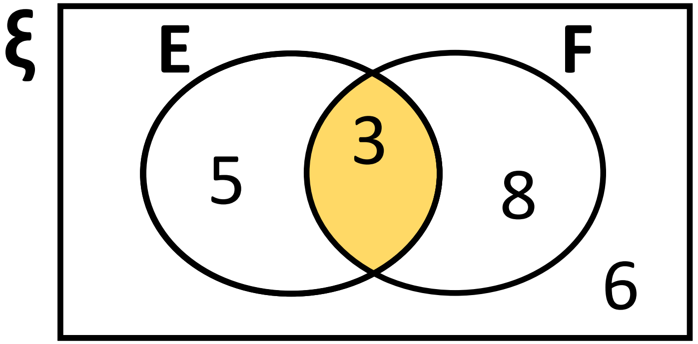
Describe & total the shaded area



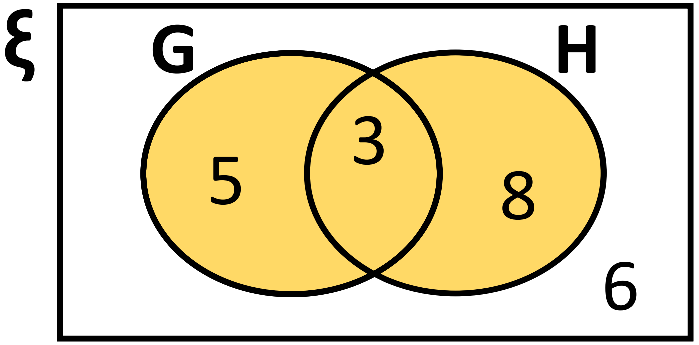
B = 11



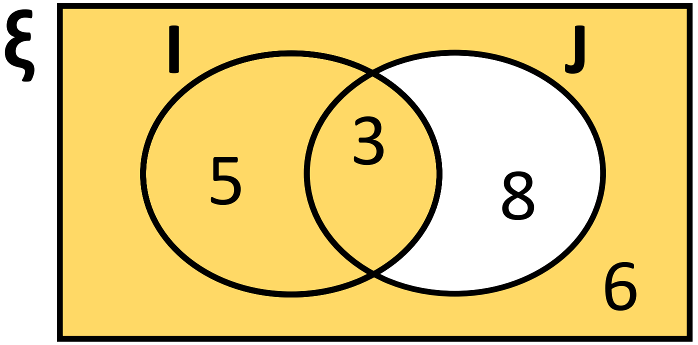
C′ = 14



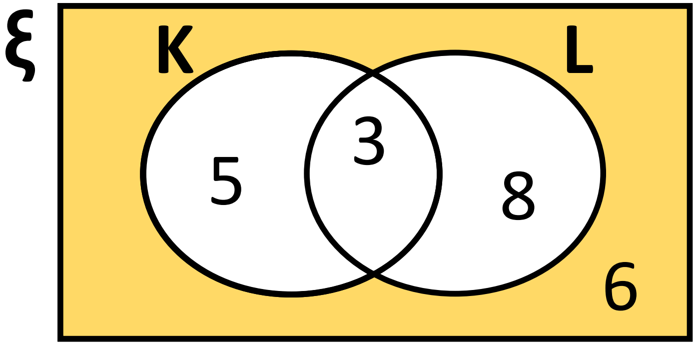
E ∩ F = 3



G ∪ H = 16



I ∪ J′ = 14



K′ ∩ L′ = 6

Example

Samples of polycarbonate plastic are analyzed for scratch and shock resistance. The results from 50 samples are summarized as follows

		shock resistance	
		high	low
scratch resistance	high	40	4
	low	1	5

Let

A = event that a sample has high shock resistance.

B = event that a sample has high scratch resistance.

		shock resistance	
		high	low
scratch resistance	high	40	4
	low	1	5

Find followings

1. $\boldsymbol{A \cap B}$
2. $\boldsymbol{A \cup B}$
3. $\boldsymbol{A'}$ (Sample in which shock resistance is low)

SOLUTION :-

$\boldsymbol{A \cap B}$= 40 (Scratch and shock resistance are high)

$\boldsymbol{A \cup B}$ = 45 (Shock resistance, scratch resistance, or both are high)

	shock resistance
scratch resistance	high
	40
	1

$\boldsymbol{A' = 9}$

Sample in which shock resistance is low

Mutually exclusive events

If the sets corresponding to events A and B are disjoint, i.e., $\boldsymbol{A} \cap \boldsymbol{B} = \emptyset$this means that they cannot happened at same time.

EXAMPLE: Consider the following two events:

A — a randomly chosen person has blood type A, and

B — a randomly chosen person has blood type B.

In rare cases, it is possible for a person to have more than one type of blood flowing through his or her veins, but for our purposes, we are going to assume that each person can have only one blood type. Therefore, it is impossible for the events A and B to occur together.

Events A and B are Disjoint

The Concept of Probability

In any random experiment there is always uncertainty as to whether a particular event will or will not occur. As a measure of the chance, or probability, with which we can expect the event to occur, it is convenient to assign a number between 0 and 1. If we are sure or certain that the event will occur, we say that its probability is 100% or 1, but if we are sure that the event will not occur, we say that its probability is zero

Classical Approach

If an event can occur in **"m"** different ways out of a total number of **"n"** possible ways, all of which are equally likely, then the probability of the event is $\frac{\boldsymbol{m}}{\boldsymbol{n}}$.

The Axioms of Probability

Axiom 1. For every event A in the class C, $\boldsymbol{P(A) \geq 0}$

Axiom 2: For the sure or certain event S in the class C, $\boldsymbol{P(S) = 1}$

Axiom 3: For any number of mutually exclusive events **(disjoint)** A_1, A_2,….

$\boldsymbol{P(A_1 \cup A_2 \cup \cdots .) = P(A_1) + P(A_2) + \cdots}$

In particular, for two mutually exclusive events A_1, A_2,

$\boldsymbol{P(A_1 \cup A_2) = P(A_1) + P(A_2)}$

Axiom 4: For any event A, $\boldsymbol{P(A) + P(A') = 1}$

$$\boldsymbol{P(A) = 1 - P(A')}$$

EXAMPLE

A statistics class for engineers consists of 25 industrial, 10 mechanical, and 10 electrical and 8 civil engineering students. If a person is randomly selected by the instructor to answer a question, find the probability that the student chosen is (a) an industrial engineering major and (b) a civil engineering or an electrical engineering major**.**

Solution

Denote by I, M, E, and C the students majoring in industrial, mechanical, electrical, and civil engineering, respectively. The total number of students in the class is 53, all of whom are equally likely to be selected

a. $P(I) = \mathbf{\frac{25}{53}}$

b. Since 18 of the 53 students are civil or electrical engineering majors, it follows that

$P(C \cup E) = \frac{18}{53}$.

EXAMPLE

A random experiment can result in one of the outcomes {a, b, c, d} with probabilities 0.1, 0.3, 0.5, and 0.1, respectively. Let A, B, and C are the events defined as

A = {a, b}, B = {b, c, d}, C = {d}

Find following $P(A)$, $P(B)$, $P(C)$, $P(A')$, $P(B')$, $P(C')$, $P(A \cap B)$, $P(A \cup B)$

Solution

As

$$P(a) = 0.1, \quad P(b) = 0.3, \quad P(c) = 0.5, \quad P(d) = 0.1$$

$$P(A) = 0.1 + 0.3 = 0.4$$

$$P(B) = 0.3 + 0.5 + 0.1 = 0.9$$

$$P(C) = 0.1$$

$$P(A') = 1 - P(A) = 1 - 0.4 = 0.6$$

$$P(B') = 1 - P(B) = 1 - 0.9 = 0.1$$

$$P(C) = 1 - P(C) = 1 - 0.1 = 0.9$$

$(A \cap B) = \{b\}$, $P(A \cap B)$= 0.3

$(AUB) = \{a, b, c, d\}$, $P(AUB)$$= 0.1 \ + 0.3 + 0.5 + 0.1 = 1$

Additional Rule 2:

When two events, A and B, are non-mutually exclusive, the probability that A or B will occur is:

$$P(A\ or\ B) = P(A) + P(B) - P(A\ and\ B)$$

EXAMPLE 2

Visual inspection of a location on wafers from a semiconductor manufacturing process resulted in the following table

Number of Contamination Particles	Proportion of Wafers
0	0.40
1	0.20
2	0.15
3	0.10
4	0.05
5 or more	0.10

i. If one wafer is selected randomly from this process and the location is inspected, what is the probability that it contains no particles?

Sol.

$$P(E) = 0.4$$

i. What is the probability that a wafer contains three or more particles in the inspected location?

Sol.

Let E denote the event that a wafer contains three or more particles in the inspected location. Then, E consists of the three outcomes {3, 4, 5 or more}. Therefore

$$P(E) = 0.10 + 0.05 + 0.10 = 0.25$$

EXAMPLE 3

Samples of emissions from three suppliers are classified for conformance to air-quality specifications. The results from 100 samples are summarized as follows:

		conforms	
		yes	no
	1	22	8
supplier	2	25	5
	3	30	10

Let A denote the event that a sample is from supplier 1, and let B denote the event that a sample conforms to specifications. If a sample is selected at random, determine the following probabilities:

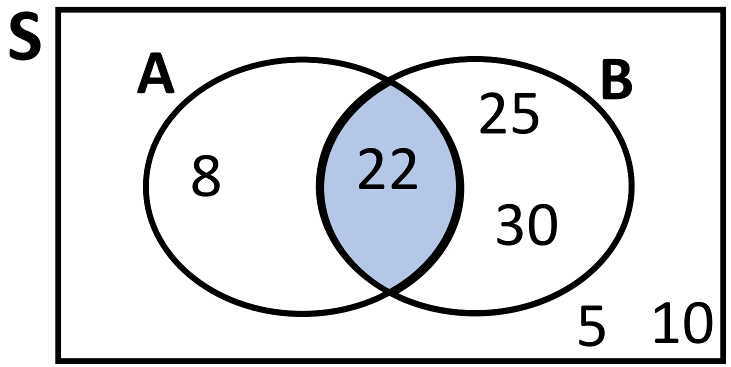
(a) $P(A)$ (b) $P(B)$
(c) $P(A')$ (d) $P(A \cap B)$
(e) $P(A \cup B)$ (f) $P(A' \cup B)$

A = Sample from supplier 1 **B** = sample confirm to specification

Intersection

A ∩ B $= 22$

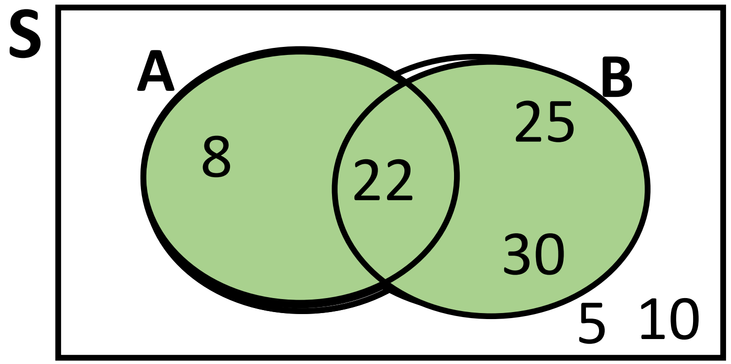
The overlap of **A** and **B**.



Intersection

A ∪ B $= 8+22+25+30=85$

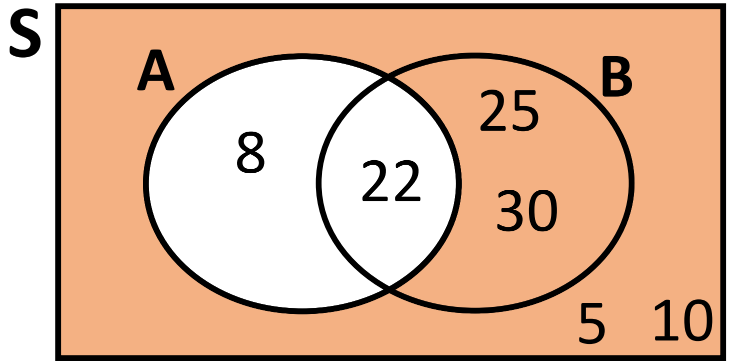
The combination of **A** and **B**.



Complement

A’ $= 25+30+5+10=70$

Not in **A**.



$$Total = 22 + 25 + 30 + 8 + 5 + 10 = 100$$

$$N(A) = 22 + 8 = 30$$

$$P(A) = \frac{N(A)}{Total} = \frac{30}{100} = 0.30$$

b. $N(B) = 22 + 25 + 30 = 77$

$$P(B) = \frac{N(B)}{Total} = \frac{77}{100} = 0.77$$

$$A' = \text{Supplier is not from sample 1} = 25 + 30 + 5 + 10 = 70$$

$$P(A') = \frac{N(A')}{Total} = \frac{70}{100} = 0.70$$

$A \cap B$ = Supplier 1 and conferm yes

$N(A \cap B) = 22$

$$P(A \cap B) = \frac{N(A \cap B)}{Total} = \frac{22}{100} = 0.20$$

$A \cup B$ = Supplier 1 Or conferm yes

$$N(A \cup B) = \mathrm{N(A)} + \mathrm{N(B)} - \mathrm{N(A \cap B)} = 30 + 77 - 22 = 85$$

=

$$P(A \cup B) = \frac{N(A \cup B)}{Total} = \frac{85}{100} = 0.85$$

2-27. Samples of a cast aluminum part are classified on the basis of surface finish (in microinches) and edge finish. The results of 100 parts are summarized as follows:

		edge finish	
		excellent	good
surface	excellent	80	2
finish	good	10	8

(a) Let A denote the event that a sample has excellent surface finish, and let B denote the event that a sample has excellent edge finish. Determine the number of samples in $A' \cap B$, B', and $A \cup B$.

EXAMPLE

The wafers such as classified as either in the “center” or at the “edge” of the sputtering tool that was used in manufacturing, and by the degree of contamination. Table below shows the proportion of wafers in each category history of 940 wafers in a semiconductor manufacturing process.

	Location in Sputtering Tool		
Contamination	Center	Edge	Total
Low	514	68	582
High	112	246	358
Total	626	314	

Suppose one wafer is selected at random find the probability that

i. wafer contains high levels of contamination

ii. Wafer is in the center of a sputtering tool.

iii. wafer is from the center of the sputtering tool and contains high levels of contamination

iv. wafer is from the center of the sputtering tool or contains high levels of contamination (or both)

Solution

i. $P(H) = \frac{358}{940}$

ii. $P(C) = \frac{626}{940}$

iii. $P(H \cap C) = \frac{112}{940}$

iv. $P(HUC) = P(H) + P(C) - P(H \cap C) = \frac{872}{940}$

1. Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows

		shock resistance	
		high	low
scratch	high	70	9
resistance	low	16	5

Let A denote the event that a disk has high shock resistance, and let B denote the event that a disk has high scratch resistance. If a disk is selected at random, determine the following probabilities:

(a) $P(A)$ (b) $P(B)$
(c) $P(A')$ (d) $P(A \cap B)$
(e) $P(A \cup B)$ (f) $P(A' \cup B)$